CLASSIFICATION OF MUSHROOM EDIBILITY

PREDICTING SAFETY THROUGH AUTOMATED ANALYSIS

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INTRODUCTION

- Mushroom classification is **critical for food safety**, as some species are highly poisonous and difficult to identify manually
- Traditional identification methods rely on expert knowledge and visual inspection of morphological features
- Manual classification is **time-consuming, subjective**, and prone to human errors
- Machine learning provides an automated approach for accurate and efficient classification
- This project focuses on predicting mushroom edibility using **structured morphological data**

LITERATURE REVIEW: RECENT ADVANCES (2025)

JACOB ET AL. **2025**

Comprehensive review of AI applications in mushroom cultivation, breeding, and classification.

QI ET AL. **2025**

Developed a Mamba YOLO-based model for detecting and classifying Shiitake mushroom fruiting bodies.

KERTA ET AL. **2025**

Improved deep learning-based mushroom image classification accuracy by optimizing techniques.

SINGH ET AL. **2025**

Proposed optimized DenseNet architectures for precise classification of edible and poisonous mushrooms.

XU ET AL. **2025**

Introduced an improved ShuffleNetV2 model for efficient and lightweight wild mushroom classification.

DATASET OVERVIEW

SOURCE

The dataset originates from the **UCI Machine Learning Repository**. It is a classic benchmark dataset for categorical classification tasks.

SIZE & FEATURES

Contains **8,124 samples** with **22 categorical features** covering various mushroom morphological traits.

TARGET VARIABLE

The objective is **Binary Classification**: distinguishing between **Edible (e)** and **Poisonous (p)** mushrooms.

DATA QUALITY

The dataset exhibits high integrity with **no missing values**, making it an ideal candidate for supervised learning benchmarks.

UCI Repository: archive.ics.uci.edu/ml/datasets/mushroom

Kaggle Dataset: kaggle.com/datasets/uciml/mushroom-classification

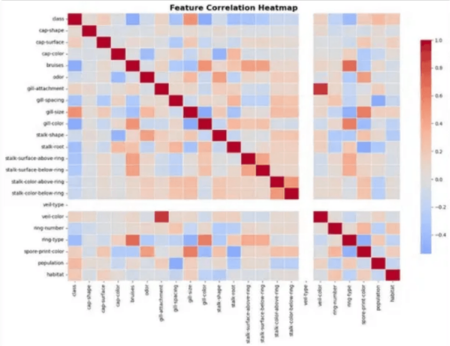
ETHICAL IMPLICATIONS

- **Data Privacy:** The dataset contains no personal or sensitive information; it consists only of mushroom morphological characteristics.
- **Open Access:** Data is publicly available via Kaggle/UCI under an open-access license permitting academic use and research.
- **Anonymization:** The dataset is fully anonymized, ensuring no individual or private entity can be identified from the morphological data.
- **Compliance:** The project strictly follows the dataset's usage license and terms of consent provided on the official repository websites.
- **Security:** All data and models are stored securely within a protected development environment to ensure integrity and prevent unauthorized access.

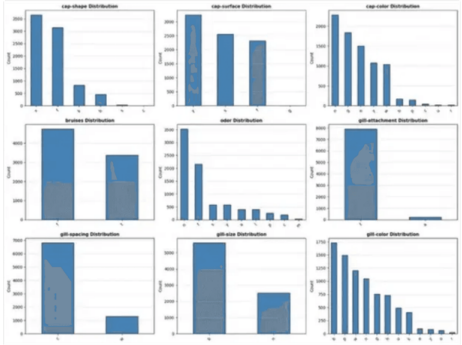
EXPLORATORY DATA ANALYSIS (EDA)

- Class distribution (edible vs poisonous) was examined to check for data imbalance
- Frequency analysis of categorical features such as cap shape, color, and odor was conducted
- Relationships between features and the target variable were analyzed to identify important predictors
- Visualizations such as count plots and distribution graphs were used to gain insights and support model building

FEATURE CORRELATION HEATMAP

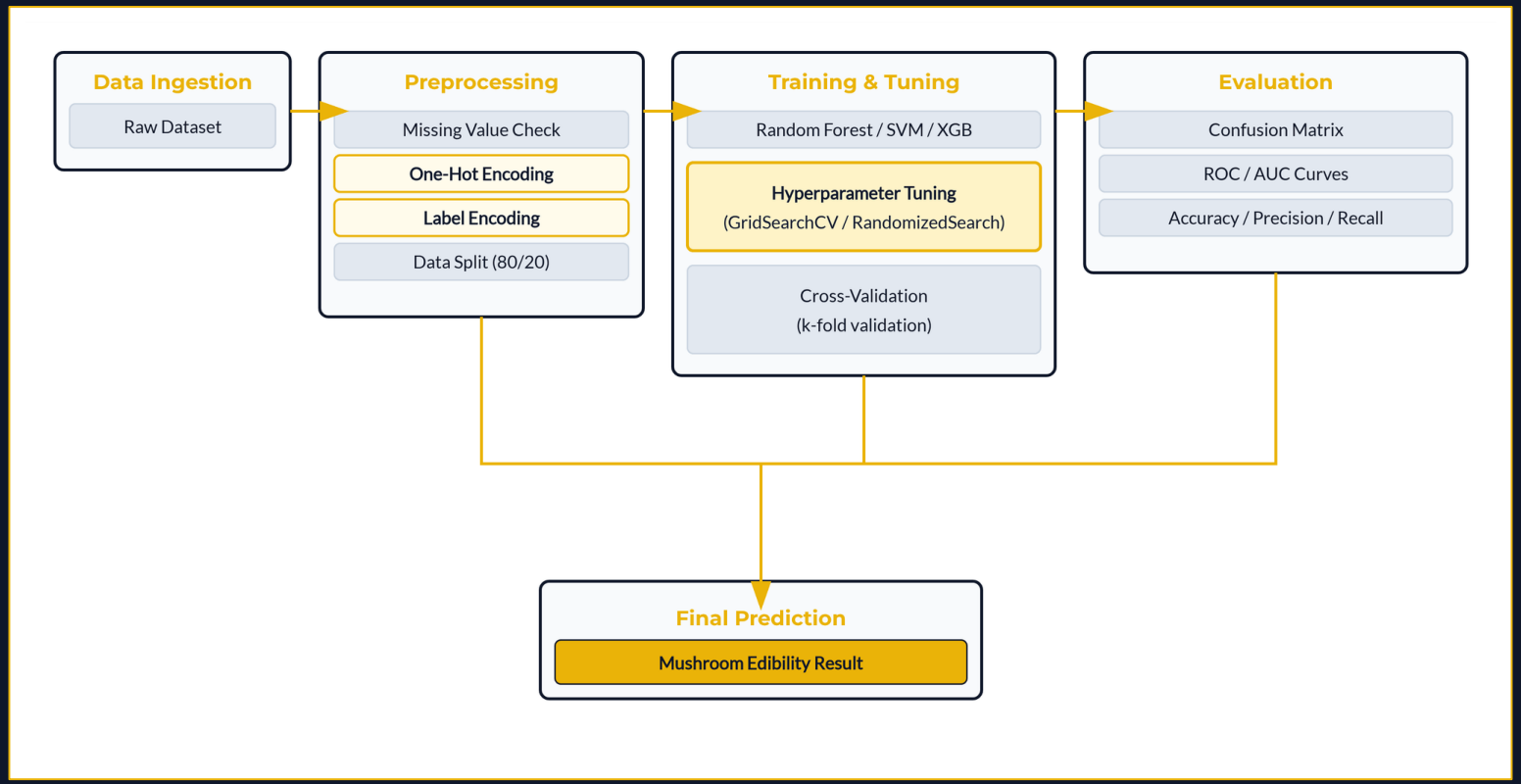


CATEGORICAL FEATURE DISTRIBUTIONS



DETAILED MODEL ARCHITECTURE

Technical workflow illustrating data transformation, model tuning, and evaluation processes.



DATA PREPROCESSING

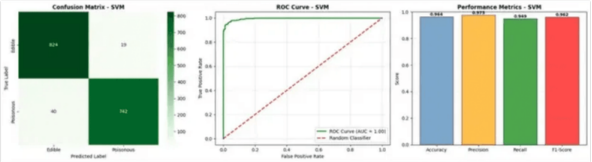
- The dataset was checked for **missing values** and found to be complete, requiring no imputation
- Categorical input features were transformed using **one-hot encoding** for machine learning compatibility
- The target variable (edible/poisonous) was converted into numerical form using **label encoding**
- The dataset was split into training and testing sets (**80% training and 20% testing**)
- Preprocessed data was used to ensure efficient training and accurate model evaluation

MACHINE LEARNING MODELS

- **Random Forest**: An ensemble learning algorithm that constructs multiple decision trees during training. It improves classification accuracy and controls over-fitting by averaging the results of individual trees.
- **SVM**: Support Vector Machine is a robust classifier that identifies the optimal hyperplane to separate classes in high-dimensional space, maximizing the margin between different categories.
- **XGBoost**: An optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable. It implements machine learning algorithms under the Gradient Boosting framework.
- Implementation includes both **raw and hyperparameter-tuned variants** for all models to ensure comprehensive performance comparison.

MODEL RESULTS & EVALUATION

SVM PERFORMANCE ANALYSIS: CONFUSION MATRIX, ROC CURVE, AND METRICS



- Models evaluated using **Confusion Matrices and ROC curves** to ensure high precision and recall
- High classification accuracy across all models demonstrates the **robustness of morphological features**
- **Support Vector Machine (SVM)** consistently achieved the highest accuracy of ~96.37%
- **Hyperparameter tuning** significantly improved the performance of Random Forest and XGBoost

MODEL COMPARISON TABLE

MODEL VARIANT	ACCURACY	PRECISION	RECALL	F1-SCORE
Random Forest (Raw)	0.7680	0.9451	0.5499	0.6952
SVM (Raw)	**0.9637**	**0.9750**	**0.9488**	**0.9618**
XGBoost (Raw)	0.9095	1.0000	0.8120	0.8963
Random Forest (Tuned)	0.9268	1.0000	0.8478	0.9176
SVM (Tuned)	**0.9637**	**0.9750**	**0.9488**	**0.9618**
XGBoost (Tuned)	0.9465	1.0000	0.8887	0.9411

SUMMARY & CONCLUSIONS

- The project successfully developed a machine learning framework to classify mushroom edibility with **high precision and reliability**.
- **Support Vector Machine (SVM)** emerged as the most accurate model, achieving a peak performance of **96.37%**.
- Morphological features such as **odor, gill-color, and spore-print-color** were identified as key predictors for edibility.
- Automated classification offers a **scalable and objective alternative** to manual identification, significantly reducing human error.
- Future work could involve integrating **computer vision techniques** to classify mushrooms directly from field images.

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THANK YOU

QUESTIONS & DISCUSSION

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